Abhay Mall

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Summary

Aspiring Software Engineer with strong foundations in **C/C++, Python, JavaScript, and Data Structures & Algorithms**. Experienced in building **backend services, REST APIs, full-stack applications, authentication systems, and database-driven solutions** using Node.js, Express.js, MongoDB, MySQL, and Redis. Hands-on experience with the **MERN stack, JWT authentication, RAG, embeddings, vector search, and LLM applications**, with a focus on clean, efficient, and maintainable code.

Technical Skills

Languages	C, C++, Java, Python, JavaScript
Backend	Node.js, Express.js, FastAPI, REST APIs, JWT, Authentication, Authorization
Frontend	React, HTML, CSS
Databases	MongoDB, MySQL, Redis, SQL, Mongoose, Database Design, Indexing
AI / LLM	RAG, Embeddings, FAISS, LLM APIs, Prompt Engineering, LangChain
Python Development	Type Hints, Async Programming, Clean Code, API Testing
CS Fundamentals	Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Tools	Git, GitHub, Postman, VS Code, npm, Docker Basics

Projects

Full-Stack Job Application Platform **2026**

MongoDB, Express.js, React, Node.js, JWT, REST APIs

- Developed a **MERN-based job platform** for job postings, discovery, applications, and recruiter management.
- Implemented **JWT authentication and role-based authorization** with protected API endpoints for candidates and recruiters.
- Designed MongoDB schemas and **REST APIs** with validation, middleware, centralized error handling, and CRUD operations.

Document Intelligence RAG Assistant **2026**

Python, FastAPI, LangChain, Gemini API, FAISS, Embeddings

- Built a **RAG-based document assistant** to answer questions using context retrieved from uploaded documents.
- Implemented **document chunking, embeddings, vector storage, and FAISS similarity search** to retrieve relevant information.
- Developed **FastAPI REST endpoints** for document uploads and question answering, using retrieved context to improve response relevance.

Distributed Rate Limiter **2026**

C++17, Redis, Multithreading, Data Structures

- Built a **distributed API rate limiter in C++** using Redis and the token bucket algorithm to control request traffic.
- Implemented **atomic Redis operations and key expiration** to maintain consistent request limits across server instances.
- Added **concurrent request handling** and separated rate-limiting logic from request processing for backend integration.

Education

2023 – Present **Bachelor of Engineering in Computer Science and Engineering**
Chandigarh University, Mohali — **CGPA: 8.27/10**

2021 **Intermediate (Class 12)**
Mahatma Gandhi Inter College, Gorakhpur — **CGPA: 7.79/10**

2019 **High School (Class 10)**
Mahatma Gandhi Inter College, Gorakhpur — **CGPA: 9.11/10**

Achievements and Certifications

- Solved **700+ coding problems** across multiple platforms, including **450+ on LeetCode (Rating: 1594)**.
- **Winner – Tekathon 2025**, Chandigarh University; secured first place in the **Smart Education** category.
- Completed **NPTEL Cloud Computing** and **Full-Stack Development certifications in Angular and Spring Boot**.